

The Core Spec (Technical Readiness)

Goal: To prove "Shelf-Ready" capability by mapping academic coursework to 2026 industry standards.

Recruiters don't know what "CS 452" means. Your job is to translate your BYU coursework into the "High-Signal" keywords that their ATS is scanning for. Below is the **BYU-to-Industry Translation Matrix** for each major emphasis.

A. The Core "Table Stakes" (Required for All Majors)

Every CS student, regardless of emphasis, must list these to pass the baseline "Competence Risk" filter.

- **Languages:** Python, C++, Java (from **CS 240: Advanced Programming Concepts**).
- **Core Concepts:** Data Structures & Algorithms (from **CS 235 & CS 312: Algorithm Design & Analysis**), Object-Oriented Design.
- **The "Sacrifice":** Do **not** list Microsoft Office, Windows, or "Typing 80 WPM." These are assumed and dilute your signal.

B. Emphasis-Specific "Power Skills" (The Differentiators)

Select your track below and ensure these specific keywords appear in your "Skills" section.

1. Software Engineering Emphasis (The Generalist Prime)

- **Source Courses:** **CS 340 (Software Engineering)**, **CS 329 (Testing, DevOps, & Agile)**, **CS 428 (Software Engineering II)**.
- **High-Signal Keywords:**
 - **Testing:** Jest, JUnit, Pytest, End-to-End (E2E) Testing (from **CS 329**).
 - **DevOps:** CI/CD Pipelines, GitHub Actions, Docker (from **CS 329**).
 - **Architecture:** RESTful APIs, Microservices, Design Patterns (Factory, Singleton).
- **Resume Win:** "Architected a CI/CD pipeline using GitHub Actions to automate testing for a Java Microservice (CS 329)."

2. Machine Learning & Data Science Emphasis

- **Source Courses:** **CS 270 (Intro to Machine Learning)**, **CS 470 (Artificial Intelligence)**, **CS 472 (Machine Learning)**, **CS 474 (Deep Learning)**.
- **High-Signal Keywords:**
 - **AI Engineering:** RAG (Retrieval-Augmented Generation), LLM Fine-Tuning, Vector Databases (Pinecone/Milvus).
 - **Libraries:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy.

- **Theory:** Deep Learning, Neural Networks, Computer Vision (from **CS 450**).
- **Resume Win:** "Built a RAG-based document question-answering system using PyTorch and a Vector Database, reducing query hallucinations by 40%."

3. Animation & Games Emphasis (The Technical Artist)

- **Source Courses:** **CS 355 (Graphics & Image Processing)**, **CS 455 (Computer Graphics)**, **CSANM 354 (Scripting for Animation)**.
- **High-Signal Keywords:**
 - **Tools:** Houdini (VEX), Maya (Python/PyMel), Blender, Unreal Engine 5 (Blueprints/C++).
 - **Graphics:** OpenGL, GLSL Shaders, Ray Tracing, RenderMan.
 - **Pipeline:** USD (Universal Scene Description), Asset Optimization.
- **Resume Win:** "Developed a custom Maya Python tool to automate character rigging, saving artists 15+ hours per project (CSANM 450)."

4. Bioinformatics Emphasis

- **Source Courses:** **BIO 465 (Capstone in Bioinformatics)**, **CS 472 (Machine Learning)**.
- **High-Signal Keywords:**
 - **Domain:** Genomic Sequencing, Protein Folding, Biological Data Analysis.
 - **Tech Stack:** R (Language), Bioconductor, Python (BioPython).
 - **Data:** Big Data Processing, Statistical Modeling (from **STAT 201**).
- **Resume Win:** "Processed 50GB+ of genomic sequencing data using Python scripts to identify protein folding anomalies."

C. The "Modernity Check" (2026 Flags)

Even if you learned it in class, phrase it like a pro.

- *Old Way:* "Learned about Databases in **CS 452 (Database Systems)**."
- *New Way:* "Designed normalized SQL schemas and optimized queries for large-scale data retrieval (PostgreSQL)."
- *Old Way:* "Used Git for class projects."
- *New Way:* "Managed version control and branch protection rules for a 5-person team using Git/GitHub."
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